



An experimental and numerical study of heave added mass and damping of horizontally submerged and perforated rectangular plates

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ABSTRACT

Forced harmonic heave motions of horizontally submerged and perforated rectangular plates are studied experimentally and numerically at both a deep and shallow submergence. The steady-state vertical forces are expressed in terms of added mass and damping coefficients. The numerical results are partly obtained by combining potential flow with linear free-surface conditions and a nonlinear viscous pressure loss condition at the mean oscillatory plate position. A domain decomposition technique is applied with a boundary element method in the inner domain and an analytical representation of the velocity potential in the outer domain. A drag term accounts for the vortex shedding at the outer plate edges. The numerically predicted Keulegan–Carpenter number dependent heave added mass and damping coefficients agree reasonably with experimental values, in particular for the deeper submergence.

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1. Introduction

Rectangular perforated templates are extensively used in the sub-sea oil fields as protection covers to shelter the sub-sea oil-field equipment from trawling and falling objects. These perforated templates are generally very heavy with large horizontal dimensions (e.g. 20 m by 20 m). During installation, when the template is being lowered from a surface crane vessel all the way down to the sea floor, large oscillatory heave accelerations of the lowering template may occur due to the wave induced heave motions of the crane vessel. The resulting vertical hydrodynamic force on the perforated template is a determining factor for structural failure of the hoist system.

Because of the complexity of the shapes of these protection templates as well as the perforations, model tests are commonly used to study their hydrodynamic properties. Molin and Nielsen (2004) conducted both forced heave and free decay model tests of scaled protection templates when deeply submerged beneath the free surface. The perforation ratio, defined as the ratio between the total projected areas of the perforation to the total projected area of the structure, was from 15% to 35%. Heave amplitude dependent added mass and damping were reported. Later Sandvik et al. (2006) performed the same kind of experiments with two types of scaled protection templates. One type was made of parallel round steel bars and the other one was made of steel grillages. The perforation ratios ranged from 15% to 47%. The dependence of added mass and damping coefficients on the heave amplitude and the perforation ratio was addressed. However, both above-mentioned experiments were carried out when the tested structures were far away from both the free surface and the bottom. In order

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to examine the free-surface effects, [Molin et al. \(2008\)](#) performed forced harmonic model tests in heave of a horizontal circular thin disk (diameter 600 mm $\times$ thickness 1 mm) with the two perforation ratios 10% and 20%. The submergences were 50 mm, 100 mm and 250 mm and the constant water depth was 500 mm. The Keulegan–Carpenter number $KC = \pi A/a$, where A is the heave amplitude and a is the disk radius, ranged from 0.025 to 1.0. They emphasized based both on numerical and experimental results that the vortex shedding from the outer edges of the disk is important for both the heave added mass and damping for the largest KC numbers. The free-surface effects were also demonstrated.

[Chua et al. \(2005\)](#) performed forced heave tests of 400 mm by 400 mm and 10 mm thick solid/perforated square plates. The application in mind was perforated plates at the lower parts of tension leg platforms (TLPs) and Spar buoys, which enhance the heave added mass and introduce extra damping ([Downie et al., 2000](#)). Chua et al. perforated the plate by cutting a single opening in the middle of the plate. The area cut from the solid plate was 180 mm by 180 mm to make a total perforation ratio of 20.25% in one case, while a 40 mm $\times$ 40 mm opening from another solid plate was cut to make a perforation ratio of 1%. They showed that the drag and added mass coefficients of these solid and perforated plates were significantly dependent on the KC number. Further, the larger the perforation ratio was, the larger the value of the drag coefficient and the smaller the value of added mass were. [Tao and Dray \(2008\)](#) tested circular solid/perforated disks (diameter 400 mm $\times$ thickness 8 mm) when deeply submerged in water. The disks were perforated by drilling a large number of small rounded openings. The perforation ratios were from zero (solid) to 20%. The KC number in their tests ranged from 0.2 to 1.2. Their experimental heave added mass and damping coefficients showed a nearly linear dependence on the KC number.

The KC number dependence of added mass and damping of the above-mentioned perforated structures has in the literature numerically been examined by using a nonlinear pressure loss condition at the plate, which expresses the pressure loss/drop through the perforation as a pressure loss/drop coefficient times the square of the cross-flow velocity. Darcy's law that provides a linear relation between the pressure loss/drop and the cross-flow velocity has also been applied by some researchers to study the wave interactions with porous media. A literature review was given by [Chwang and Chan \(1998\)](#). Even though Darcy's law is numerically convenient, it is believed to be a less adequate description of the pressure loss condition in our study i.e. for perforated plate of small thickness with clearly open areas. That is because if the perforated openings are sharp-edged and the size is large enough for the flow to separate, the pressure forces dominate the frictional forces and Darcy's law is not suitable.

For the nonlinear pressure loss/drop model, the pressure loss/drop coefficient has to be experimentally determined and in the case of steady ambient flow it depends on the geometrical structure of the perforations, Reynolds number, perforation ratio, and the angle between the approaching flow to the perforated body surface normal vector ([Blevins, 1984](#)). When the body oscillates and/or there is an unsteady ambient cross-flow, the KC number must also be considered. [Molin and Legras \(1990\)](#) first used the above-mentioned nonlinear pressure loss model in combination with a potential flow to study the hydrodynamic properties of a slotted circular cylinder that was used as a stabilizer for a compliant jacket platform. Successful results were reported and it was shown that both the added mass and damping coefficients of the slotted cylinder depended on the amplitude, i.e. the KC number. The same approach has been applied afterwards to a variety of problems and geometries like waves propagating through a series of perforated vertical plates ([Molin and Fourest, 1992](#)), circular cylinder with a concentric thin perforated circular shrouds in either uniform or oscillatory flow ([Molin, 1993](#)), periodic arrays of parallel perforated disks in an unbounded fluid domain ([Molin, 2001](#)), a horizontally submerged and perforated disk heaving with or without free-surface and sea-floor effects ([Molin and Nielsen, 2004](#); [Molin et al., 2008](#)), and water entry of a perforated wedge ([Molin and Korobkin, 2001](#)). When the KC number was relatively small, all the experimental and theoretical results for the perforated body studied in the above-mentioned papers were in good agreement. Further, the nonlinear pressure drop model well explained the motion-amplitude dependence of the heave added mass and damping coefficients. Only one exception is for the perforated shrouded cylinder case. The main problem in the latter case is failure in correctly reproducing the flow in the downstream part of the annular region, which is between the shroud and the non-perforated cylinder, even for the case of very small perforations.

The perforated body geometries in the above numerical studies were simple, and by taking advantage of the symmetry properties of both the flow and the body, most of the analyses were carried out in a semi-analytical way, and some problems were solved in 2D. In our following numerical calculations, we will still utilize the main concept of this nonlinear pressure drop model and combine it with a potential flow with bottom effects and linear free-surface conditions to study the hydrodynamic behavior of heaving horizontally submerged and perforated 3D rectangular plates. A domain decomposition method is applied with a boundary element method (BEM) in the inner domain and an analytical representation of the velocity potential in the outer domain. One advantage of using a BEM to study the perforated plate is no restrictions to the shapes of the plate. We may also account for other motion modes and the effect of incident waves. Further, the BEM could be easily implemented and applied to study the flow caused by the perforated plate together with other bodies of interest (e.g. damping plates of TLPs and Spar buoys, a perforated plate being lowered from a surface crane vessel). Using a CFD solver based on Navier–Stokes equations to solve the detailed flow through the many perforations in our cases would be CPU-demanding.

When a solid plate of finite thickness with sharp edges oscillates in water, vorticity will also be shed from the outer edges, which will change the pressure acting on the plate and affect both the added mass and damping coefficients. [Prislin et al. \(1998\)](#) examined how the added mass and drag coefficients of a solid square-plate in infinite fluid vary with the Reynolds number (the width of the plate being the characteristic length) and with the KC number. Their conclusion was that

the added mass of solid plates does not depend on the Reynolds number and that the drag coefficient at Reynolds numbers higher than 10^5 does not vary significantly with Reynolds number. However, both the added mass and drag coefficients depend on the KC number. Experimental data for solid disks in infinite fluid provided by Lake et al. (2000) show that the damping coefficients vary linearly with the KC number in the range from 0.1 to 0.5. The ranges of both the Reynolds number and the KC number in our experiments are the same as theirs. Further, both above-mentioned tests show that the Reynolds number is not a main parameter for sharp corners within the range of Reynolds number of practical importance. He et al. (2008) reported experiments of heaving circular solid disks with thickness-to-diameter ratios between 1/87.5 and 1/25, and with different KC numbers ranging from 0.01 to 1.1. From their experimental results for oscillating thin disks, the thickness-to-diameter ratio had some influence on the hydrodynamic damping. However, when $KC/2\pi$ is large relative to the thickness-to-diameter ratio, the effect of the thickness-to-diameter ratio is expected to be small. So in our numerical and experimental studies we will neglect the influence from Reynolds number and thickness effects on the drag coefficients. The length-to-breadth ratio of a rectangular plate is also a factor. However, since the considered KC numbers are small, we will neglect the influence and apply results for 2D plates.

When the plate is perforated, the water will flow both through the perforated openings as well as around the outer edges with accompanying rotational flow. Further, when the perforated plate is oscillating sufficiently close to the free surface, the damping coefficient will also be affected by the radiated surface waves.

We will start with presenting our numerical method which combines potential flow with a nonlinear pressure loss condition at the plate. The problem is solved in the frequency domain with linear free-surface conditions by using domain decomposition. New experimental results for forced harmonic heave motions of horizontally submerged and perforated rectangular plates with submergences 10 cm and 70 cm are presented. The plate length, breadth (B) and thickness are 520 mm, 365 mm, and 5 mm, respectively. The perforation ratios are 7.945% and 15.89%. The KC number $KC = 2\pi A/B$ varies between 0.17 and 1.72. Generally satisfactory agreement between numerical results and experiments is demonstrated, in particular, for the large submergence.

2. Numerical model

A numerical method has been developed to study forced small-amplitude vertical harmonic oscillations of a horizontally submerged and perforated rigid plate in the frequency domain. Since the Keulegan–Carpenter number is small, the vorticity domain due to flow separation and jet flow at the plate openings is limited to the close vicinity of the plate. The latter fact allows potential flow as a first approximation in the water domain exterior to the top and bottom plate surfaces. Flow separation will in reality occur at the outer edges of the plate. The latter fact is not considered in the numerical modeling of the flow but is added as a corrective effect in the evaluation of the hydrodynamic force by an empirical drag term. Since the water is assumed incompressible, the velocity potential is governed by the Laplace equation in the irrotational flow domain. Linear free-surface conditions without surface tension are applied and the water bottom is horizontal. Body boundary conditions are applied on the mean oscillatory plate position. The numerical procedure can be generalized to study forced roll and pitch as well as the influence of incident waves. We will refer to a rectangular plate as a consequence of later comparisons with experiments. However, the plate can have any shape.

A nonlinear pressure drop model is applied at the plate. Since the plate openings are assumed sharp-edged, the flow separates at the edges for any KC number and the pressure drop ΔP between the upper and lower solid parts of the plate can according to Molin (1992) be related to the relative velocity V_{rt} through the openings as

$$\Delta P = \frac{1}{2\mu} \rho_w V_{rt} |V_{rt}|. \quad (1)$$

Here μ is the discharge coefficient, which depends on the shape of the openings, local Reynolds number, and the angle between the approaching flow and the plate surface. Typical values of μ are in the range of 0.5–1.0 for ambient steady uniform cross-flow. ρ_w is the density of water.

The openings on the plate are assumed numerous and evenly distributed so that they can be assimilated as a uniform perforation. Under this assumption the pressure difference between the two horizontal sides of the plate is considered continuously distributed over the plate surface with resulting continuously distributed water velocities through the perforations. So the varying pressure drop ΔP and the velocity V_{rt} in (1) at each single opening are replaced by the equivalent continuous ones Δp and v_{rt} . Further, under this assumption, it makes no difference whatsoever the shape of the openings is, except for some slight changes for the discharge coefficient. We can by following Molin (1992) write the continuous pressure drop Δp on the solid parts of the plate as

$$\Delta p = \frac{1-\tau}{2\mu\tau^2} \rho_w v_{rt} |v_{rt}|, \quad (2)$$

where τ is the perforation ratio defined between 0 and 1.

A Cartesian coordinate system o - xyz is chosen with the z -axis pointing vertically upwards, and the x -axis and y -axis along the mean free surface at $z = 0$ (see Fig. 1(a)). A perforated plate submerged at $z = -d$ (d being the distance between the upper plate surface and calm free surface) with the closed boundary S_p (six surfaces of the thin plate, with upper and lower surfaces denoted as S_p^+ and S_p^- , and the rest of the four surfaces as S_p') is time-harmonically heaving with the motion

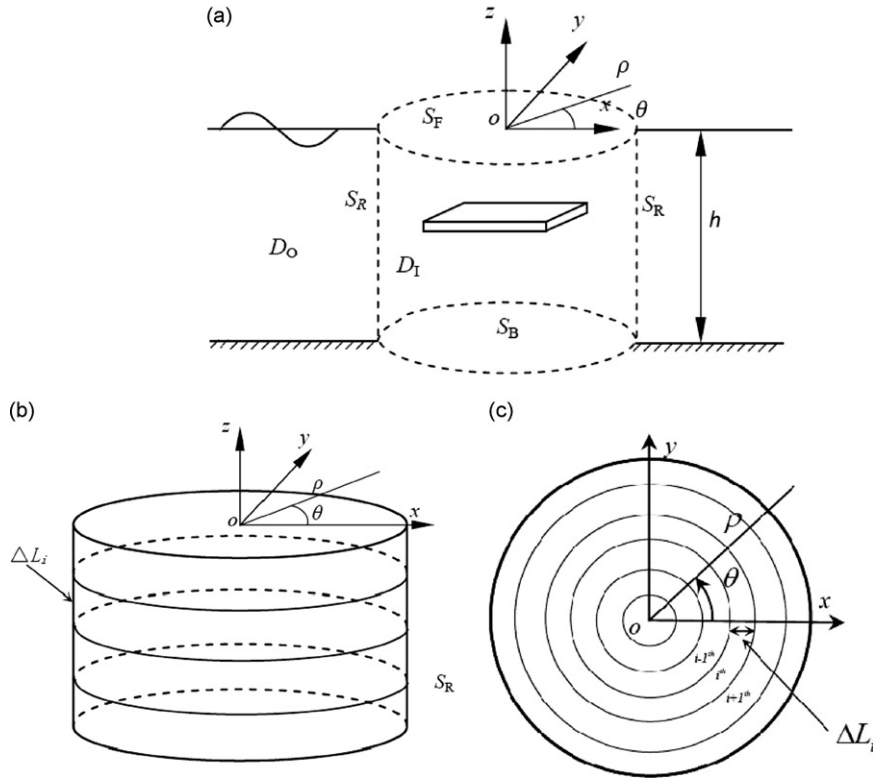


Fig. 1. (a) Definition of Cartesian coordinates together with illustration of the plate and surfaces used in a domain decomposition procedure, (b) Illustration of axis-symmetric panels on the control surface and (c) Illustration of axis-symmetric panels on the mean free surface.

$S_3 = \text{Re}[Ae^{i\omega t}]$. Since the perforations are not included in S_p , S_p^+ and S_p^- include both parts of the solid plate surface and surfaces at the perforation entrances. A is the heave amplitude, and $\omega = 2\pi/T$ is the frequency of oscillation with T as the oscillation period. Further, the velocity of the oscillating plate can be written as $V_3 = \text{Re}[iA\omega e^{i\omega t}]$. We will as a general rule use capital letters for time-dependent flow variables, and lower case characters for complex amplitudes of the flow variables. The time harmonic velocity potential is expressed as $\phi(x, y, z, t) = \text{Re}[\phi(x, y, z)e^{i\omega t}]$. Then the vertical water velocity normal to the upper and lower sides of the plate can be written as $V^+ = \text{Re}[(\partial\phi^+/\partial n^+)e^{i\omega t}]$ on S_p^+ , $V^- = \text{Re}[(\partial\phi^-/\partial n^-)e^{i\omega t}]$ on S_p^- . Here ϕ^+ and ϕ^- are the complex velocity potentials and n^+ and n^- are the surface normal vectors on S_p^+ and S_p^- , respectively. The positive normal direction is pointing from the water into the plate surfaces. Due to the conservation of fluid mass through the openings, we have $V^+ = -V^-$. Then the relation between the pressure difference and relative cross-flow velocity through the openings Eq. (2) is expressed as

$$P^- - P^+ = \rho_w \frac{1-\tau}{2\mu\tau^2} \text{Re} \left[\left(\frac{\partial\phi}{\partial z} - iA\omega \right) e^{i\omega t} \right] \left| \text{Re} \left[\left(\frac{\partial\phi}{\partial z} - iA\omega \right) e^{i\omega t} \right] \right| \quad (3)$$

on the mean oscillatory plate position. Here P^+ and P^- are the pressures acting on the upper and lower plate surfaces, respectively. The above equation is also called the discharge equation.

Lorents' equivalent linearization

$$\text{Re} [f e^{i\omega t}] \left| \text{Re} (f e^{i\omega t}) \right| \approx \frac{8}{3\pi} \|f\| \text{Re} [f e^{i\omega t}] \quad (4)$$

is used on the right-hand side of Eq. (3) in order to separate the time harmonic terms with frequency ω . Here $f = \partial\phi/\partial z - iA\omega$, and $\| \cdot \|$ is the modulus of the complex number. The linearized Bernoulli's equation is applied on the left-hand side of Eq. (3), i.e.

$$P^- - P^+ = -\rho_w (\partial\phi^-/\partial t - \partial\phi^+/\partial t). \quad (5)$$

Then the quadratic pressure loss condition on the mean oscillatory plate position can be expressed by the following relation between the velocity potential difference and the normal velocity potential gradient

$$-i\omega(\phi^- - \phi^+) = \frac{8}{3\pi} \frac{1-\tau}{2\mu\tau^2} \left\| \frac{\partial\phi}{\partial z} - iA\omega \right\| \left(\frac{\partial\phi}{\partial z} - iA\omega \right). \quad (6)$$

The body boundary condition on the four vertical plate surfaces is zero normal velocity and is applied on the mean oscillatory position of S_p^r , i.e.

$$\frac{\partial \phi}{\partial n} = 0 \quad \text{at } S_p^r. \quad (7)$$

Further, the complex velocity potential ϕ satisfies Laplace's equation

$$\nabla^2 \phi = 0, \quad (8)$$

in the water domain outside the mean position of the perforated plate. In addition, we assume the linear free-surface condition

$$\frac{\partial \phi}{\partial z} - \frac{\omega^2}{g} \phi = 0 \quad \text{at } z = 0, \quad (9)$$

where g is the gravitational acceleration. If the submerged plate is close to the free surface, resonant water motion can occur above the plate. The latter fact requires nonlinear free-surface conditions and is not pursued in this study. Further, the water bottom condition is

$$\frac{\partial \phi}{\partial z} = 0 \quad \text{at } z = -h, \quad (10)$$

where h is the constant water depth. The radiation condition for ϕ is

$$r^{1/2} \left(\frac{\partial \phi}{\partial r} - i \frac{\omega^2}{g} \phi \right) \rightarrow 0 \quad \text{as } r = (x^2 + y^2)^{1/2} \rightarrow \infty. \quad (11)$$

The main difficulty in solving the boundary value problem described by Eqs. (6)–(11) arises from the nonlinear pressure loss condition (6), due to the fact that the velocity potential and its normal derivative are coupled. In order to overcome this difficulty, an iterative procedure is applied. First we assume an initial value of the water velocity distribution ($\partial \phi / \partial z$ -term in Eq. (6)) on the upper and lower perforated plate boundary. By solving the resulting boundary value problem once, we find the velocity potential (dynamic pressure) on S_p^+ and S_p^- . By evaluating the $\|\partial \phi / \partial z - iA\omega\|$ -terms from the initial value of velocity distribution on the right-hand side of Eq. (6) together with the new ϕ^+ and ϕ^- on the left-hand side, we find an updated velocity distribution. If the relative difference between the previous water velocity distribution and the updated one is less than a small prescribed value, convergence is achieved.

The surface of a vertical cylinder of radius R with axis along the z -axis that extends from the sea floor to the mean free surface is used as a control surface in the numerical method. We let the radius R be sufficiently large to include the perforated plate inside the control surface. The control surface divides the water domain into inner and outer sub-domains D_1 and D_0 (see Fig. 1(a)). $\phi^i e^{i\omega t}$ and $\phi^o e^{i\omega t}$ are the velocity potentials in D_1 and D_0 , respectively. In the outer fluid domain D_0 the velocity potential $\phi^o e^{i\omega t}$ can be written as the following sum of fundamental solutions of the 3-D Laplace equation (8), satisfying the linear free-surface condition (9), sea-bottom condition (10) and radiation condition (11) in the cylindrical coordinates (ρ, θ, z) as

$$\begin{aligned} \phi^o = & [C_{00} H_0^{(2)}(k_0 \rho) + \sum_{m=1}^{\infty} H_m^{(2)}(k_0 \rho) (C_{m0} \cos m\theta + D_{m0} \sin m\theta)] \cosh k_0(z+h) \\ & + \sum_{n=1}^{\infty} [C_{0n} K_0(k_n \rho) + \sum_{m=1}^{\infty} K_m(k_n \rho) (C_{mn} \cos m\theta + D_{mn} \sin m\theta)] \cos k_n(z+h). \end{aligned} \quad (12)$$

Here $H_0^{(2)}$ and $H_m^{(2)}$ are the zeroth order and m th-order Hankel function of the second kind, while K_0 and K_m are the zeroth order and m th-order modified Bessel function of the second kind, ($m = 1, 2, 3, \dots$). k_0 is the positive root of the equation $k \tanh kh = \omega^2 / g$ (h is the water depth). k_n ($n = 1, 2, 3, \dots$) are the positive roots of the equation $k_n \tan(k_n h) = -\omega^2 / g$ in the ascending order. It follows from the asymptotic expansion of the Hankel functions when $\rho \rightarrow +\infty$ and noting the fact that the evanescent mode terms $K_m \rightarrow 0$, $m = 0, 1, 2, \dots$ when $\rho \rightarrow +\infty$ that the radiation condition is satisfied. The coefficients C_{mn} ($m = 0, 1, 2, \dots$; $n = 0, 1, 2, \dots$) and D_{mn} ($m = 1, 2, 3, \dots$; $n = 0, 1, 2, \dots$) are determined together with the inner domain solution.

In the inner fluid domain D_1 we will reduce the Laplace equation for ϕ^i to a boundary integral equation by applying Green's second identity with the fundamental solution $1/r_{pq}$, where r_{pq} is the distance between the field point $p(x, y, z)$ and the source point $q(x_0, y_0, z_0)$, with p always selected outside the inner fluid domain. Then the resulting boundary integral equation is

$$\oint_{S_{\text{all}}} \left[\frac{1}{r_{pq}} \frac{\partial \phi^i(q)}{\partial n_q} - \phi^i(q) \frac{\partial}{\partial n_q} \left(\frac{1}{r_{pq}} \right) \right] ds_q = 0, \quad (13)$$

where S_{all} is the closed inner domain boundary, including parts of the mean free surface S_F , sea bottom S_B , the six plate surfaces S_P and the control surface S_R . $\partial / \partial n_q$ denotes the normal differentiation at source point q on S_{all} in the direction from the fluid domain pointing outwards. On the control surface, the matching condition is imposed, which requires the

continuity of the velocity potential and the normal velocity as

$$\phi^i = \phi^o, \quad \frac{\partial \phi^i}{\partial n} = \frac{\partial \phi^o}{\partial n}. \quad (14)$$

The boundary integral equation formulation in the inner domain requires finite plate thickness and can be solved numerically by dividing the boundaries into panels. Because the plate surface S_p is represented as six rectangular flat surfaces, quadrilateral panels are used, and on each quadrilateral panel ΔQ_i the velocity potential is assumed constant. Further, on both sides of the plate surface we use the same mesh (same panel size and horizontal position), which is convenient in calculating the pressure difference between adjacent panels on the upper and lower plate sides. Because of the axis-symmetric property of the control surface S_R , and since we have expanded the outer-domain velocity potential on the control surface (see Eq. (12)) as a function of θ , axis-symmetric panels are distributed on S_R by dividing S_R into concentric rings (see Fig. 1(b)). Axis-symmetric panels are also distributed on the mean free surface and sea-bottom boundaries in the inner domain. An illustration of the axis-symmetric panels on the mean free surface is shown in Fig. 1(c). The unknown velocity potential on each panel (concentric-ring elements) on these surfaces is expressed in a Fourier series as

$$\phi(\rho, \theta, z) = \sum_{m=0}^M \phi_{cm}(\rho, z) \cos m\theta + \sum_{m=1}^M \phi_{sm}(\rho, z) \sin m\theta. \quad (15)$$

By substituting the Fourier expansion of the velocity potential (15) on the mean free surface and sea-bottom surface into the boundary integral Eq. (13), we have the following contribution:

$$\iint_{S_F+S_B} \left[\frac{1}{r_{pq}} \frac{\partial \phi(q)}{\partial n} - \phi(q) \frac{\partial}{\partial n} \left(\frac{1}{r_{pq}} \right) \right] dS_q = \sum_{m=0}^M \cos m\theta_p \int_{\Delta L_i} \rho \left(\frac{\partial \phi_{cm}}{\partial n} - \phi_{cm} \frac{\partial}{\partial n} \right) S_m dL + \sum_{m=1}^M \sin m\theta_p \int_{\Delta L_i} \rho \left(\frac{\partial \phi_{sm}}{\partial n} - \phi_{sm} \frac{\partial}{\partial n} \right) S_m dL. \quad (16)$$

from integration over S_B and S_F . Here ΔL_i is the intersection line of the i th axis-symmetric panel (on either free surface or sea-bottom surface) and the plane $\theta = 0$. Further,

$$S_m = \int_0^{2\pi} \frac{\cos m\theta d\theta}{(a-b \cos \theta)^{1/2}}, \quad a = \rho^2 + \rho_p^2 + (z-z_p)^2, \quad b = 2\rho\rho_p, \quad (17)$$

where (ρ_p, θ_p, z_p) is the cylindrical coordinate of the field point p and (ρ, θ, z) is the coordinate of the integration point q on ΔL_i . Then the surface integral is reduced to a summation of the line integrals (see Eq. (16)). The numbers of axis-symmetrical panels on the mean free surface, sea bottom and control surface are N_{Fa} , N_{Ba} and N_R , respectively. The total number of quadrilateral panels on the perforated plate is N_p . We let $\phi_{cmi}^{S_a}$ and $\phi_{smi}^{S_a}$ denote the unknown Fourier coefficients on the axis-symmetrical panels on the free surface and sea bottom. Further, $\phi_i^{S_p}$ is the unknown velocity potential on the quadrilateral panels ΔQ_i on the plate surfaces. The unknowns on the control surface are the outer-domain velocity potential expansion coefficients C_{mn} and D_{mn} in Eq. (12). The following equation system is obtained by using the boundary conditions (Eqs. (7), (9), (10), (14)) on the surfaces ($S_p + S_F + S_B + S_R$)

$$\begin{aligned} & - \sum_{i=1}^{N_p} \iint_{\Delta Q_i} \frac{\partial}{\partial n_q} \left(\frac{1}{r_{pq}} \right) dS_q \phi_i^{S_p} + \sum_{i=1}^{N_{Fa}} \left\{ \sum_{m=1}^M \left[\sin m\theta_p \int_{\Delta L_i} \rho \left(\frac{\omega^2}{g} - \frac{\partial}{\partial n} \right) S_m dL \right] \phi_{smi}^{S_F} \right\} \\ & - \sum_{i=1}^{N_{Ba}} \left[\sum_{m=0}^M \left(\cos m\theta_p \int_{\Delta L_i} \rho \frac{\partial S_m}{\partial n} dL \right) \phi_{cmi}^{S_B} \right] - \sum_{i=1}^{N_{Ba}} \left[\sum_{m=1}^M \left(\sin m\theta_p \int_{\Delta L_i} \rho \frac{\partial S_m}{\partial n} dL \right) \phi_{smi}^{S_B} \right] \\ & + \sum_{m=0}^M \cos m\theta_p \left(\sum_{i=1}^{N_R} \int_{\Delta L_i} R U_{mi} dL \right) C_{m0} + \sum_{m=1}^M \sin m\theta_p \left(\sum_{i=1}^{N_R} \int_{\Delta L_i} R U_{mi} dL \right) D_{m0} \\ & + \sum_{n=1}^N \left[\sum_{m=0}^M \cos m\theta_p \left(\sum_{i=1}^{N_R} \int_{\Delta L_i} R U_{mi} dL \right) C_{mn} \right] + \sum_{n=1}^N \left[\sum_{m=1}^M \sin m\theta_p \left(\sum_{i=1}^{N_R} \int_{\Delta L_i} R U_{mi} dL \right) D_{mn} \right] = \sum_{i=1}^{N_p} \frac{\partial \phi_i^{S_p}}{\partial z} \cdot \vec{n} \iint_{S_p^+ + S_p^-} \frac{1}{r_{pq}} dS_q \quad (18) \end{aligned}$$

where

$$\begin{aligned} U_{mi} &= \frac{k_0}{2} \left[H_{m-1}^{(2)}(k_0 R) - H_{m+1}^{(2)}(k_0 R) \right] \cosh k_0(z+d) S_m - H_m^{(2)}(k_0 R) \cosh k_0(z+d) \frac{\partial S_m}{\partial n}, \\ V_{mi} &= -\frac{k_n}{2} \left[K_{m-1}(k_n R) + K_{m+1}(k_n R) \right] \cos k_n(z+d) S_m - K_m(k_n R) \cos k_n(z+d) \frac{\partial S_m}{\partial n}. \end{aligned}$$

On the perforated plate boundary S_p , the integration points are located at the geometrical center of quadrilateral panels. On the axis-symmetric boundaries, the numbers for the field points are the same as the unknown Fourier coefficients. Equation system (18) is solved for given vertical flow velocities $\partial \phi_i^{S_p} / \partial z$ along the perforated plate. When evaluating the

coefficients of equation system (18), four basic integrations will be encountered. The evaluation of the integral $\iint_{\Delta Q} (1/r_{pq}) dS$ and $\iint_{\Delta Q} (\partial/\partial n)(1/r_{pq}) dS$ for a quadrilateral panel is well-known (Hess and Smith, 1967). When evaluating the line integral $\int_{\Delta L} f(\rho, z) S_m dL$ and $\int_{\Delta L} f(\rho, z) (\partial S_m / \partial n) dL$ (where $f(\rho, z)$ is a known function on the intersection line ΔL_i), a 10 points Gauss–Legendre quadrature is used.

By using the earlier stated iteration procedure, convergence is achieved after several iterations of solving the boundary integral equation (18) and the discharge equation (6). The convergence speed is generally a function of the total panel numbers used in the boundary integral equation, initially imposed velocity distribution $\partial \phi_i^{Sp} / \partial z$ through the perforation, and the convergence criterion of the iteration. In our following numerical results, we distribute 3012 panels on each side of the plate surface S_p^+ and S_p^- , and 364 panels on the four thin surfaces S_p^r . For the axis-symmetric free-surface boundary S_F , control surface S_R and sea-bottom surface S_B we use 100 concentric-ring elements on each surface with $m = 8$ in the Fourier series expansion. In Eq. (12), we choose $n = 8$. The convergence criterion to control the end of the iteration is that the relative difference of averaged vertical fluid velocity along the plate surface between two successive iterations is less than 10^{-4} . In general, the convergence will be achieved after less than 30 iterations.

After the convergence is achieved a hydrodynamic force on the plate follows by integrating the pressure difference along the perforated plate surfaces. A deficiency in the presented method is that it does not account for the flow separation from the outer edges. An additional empirical drag term as in Morison equation is added and expressed as

$$F_{\text{drag}} = -\frac{1}{2} \rho C_d S (1-\tau) V_r |V_r|, \quad (19)$$

where S is the the projected area of the plate on the x – y plane. Since the drag force only acts on the solid part of the perforated plate surface, the factor $1-\tau$ is introduced in Eq. (19). The drag coefficient C_d is related to the Keulegan–Carpenter number $KC = 2\pi A/B$ in our present study (where A is the plate heaving amplitude; B is the breadth of the plate) by using 2D results for a plate without perforations at small KC number. It follows from the experimental results by Bearman et al. (1985) that

$$C_d = \alpha KC^{-1/3} \quad \text{with } \alpha = 8.0. \quad (20)$$

The flow through the perforations is related to the vortex shedding from the outer edges of the plate and thereby to which expression to use for V_r in Eq. (19). We follow the procedure suggested by Molin et al. (2008) by expressing V_r as the plate velocity minus the relative fluid velocity through the perforation averaged over the plate surface. This reference velocity V_r used in Eq. (19) equals to the vertical flow velocity averaged over the plate surface, i.e. $V_r = \text{Re}[\iint_{S_p^+} (\partial \phi / \partial z) e^{i\omega t}] / S$. As a result, this supplementary drag force does not only affect the heave damping but also the heave added mass.

3. Experiments

3.1. Experimental set-up

The experiments were performed in the Marine Cybernetics Laboratory at the Norwegian University of Science and Technology, Trondheim, Norway. The tank has a length of 40 m and a width of 10 m. The water depth is 1.40 m. The test model was located in the middle of the tank in order to minimize the reflected wave effects from the tank walls.

The two rectangular perforated plate models are made of 5 mm thick aluminum, with a length L of 520 mm and a breadth B of 365 mm. The circular plate openings are 10 mm in diameter and evenly distributed, with their edges sharpened prior to the experiments (see Fig. 2). The tested perforation ratios are 7.945% and 15.89%, corresponding to 192 and 384 holes, respectively. The experimental set up is illustrated in Fig. 3.

The vertical motion of the plate is excited by two vertically moving hydraulic actuators. Each actuator can generate controlled regular (harmonic) and irregular vertical motions. In our tests, the two actuators are linked together with a support structure and conduct the same harmonic vertical motions. A force sensor is mounted right under the support structure. A thin rod is used between the force sensor and the plate with one end connected to the sensor and the other one connected to the center of the plate. The model is horizontally submerged beneath the free surface. Two submergences $d = 70$ cm and 10 cm are studied.

Since the free-surface effects are rather small for the deep submergence case ($d = 70$ cm), only four heave periods ranging from 0.5 s to 2.5 s are tested. For the small submergence case ($d = 10$ cm), the test periods are chosen to cover the whole period range of interest for the free-surface wave generations, i.e. 14 periods ranging from 0.6 s to 3.0 s are tested. The lower limit of the test periods is due to the maximum operational limit of velocity and acceleration of the actuators. The upper limit of the test periods is chosen to have nearly zero free-surface wave generation according to the numerical predictions.

The amplitudes tested for the perforated plate ranged from 1 cm to 10 cm (for 70 cm submergence case), and 1 cm to 8 cm (for 10 cm submergence case). The corresponding KC numbers are between 0.17 and 1.72. For the small submergence case, the upper limit of the amplitudes is chosen to ensure that the plate is fully submerged below the instantaneous free surface.

Two wave gauges are used in the experiments, with one placed near the edge of the perforated plate and the other one placed close to the nearest tank wall. From the readings of the wave gauges we can choose the proper time windows for calculating the hydrodynamic coefficients to avoid the influence of the reflected waves from the tank walls. Two accelerometers are placed on the surface of the plate, with one right in the middle and the other one at a corner. Use of two accelerometers enabled us to check the plate rigidities.



Fig. 2. Tested perforated plate, perforation ratio $\tau = 15.89\%$, length $L = 520$ mm, breadth $B = 365$ mm, and hole diameter $D = 10$ mm.

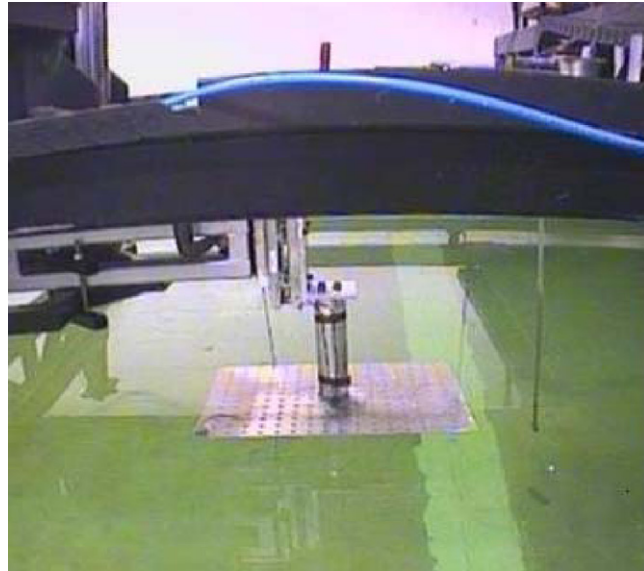


Fig. 3. Experimental set-up.

The sampling frequency for all the measuring channels is 200 Hz for all the heave periods. The filtering frequency for all the readings is set to 10 Hz.

3.2. Hydrodynamic coefficients

Steady-state forces are achieved after a few oscillations. Fig. 4 shows the typical readings of measured filtered vertical force, vertical displacement, velocity and acceleration of the perforated plate.

The noise from accelerometer 2 which is placed at a corner of the plate is probably due to small-amplitude vibrations of the model. The heave added mass A_{33} and damping coefficient B_{33} of the perforated plate for each test run are first calculated from the corresponding selected time history of the total force ($F(t)$), velocity ($\dot{Z}(t)$), acceleration ($\ddot{Z}(t)$) and displacement $Z(t)$ readings using the least square method and by expressing the force as

$$F(t) = -(M + A_{33})\ddot{Z}(t) - B_{33}\dot{Z}(t) - kZ(t). \quad (21)$$

Here M is the total structural mass including the dry mass of the thin rod and the perforated plate; k is the total waterplane area of the thin rod multiplied by the water density and the gravitational acceleration. The obtained added mass

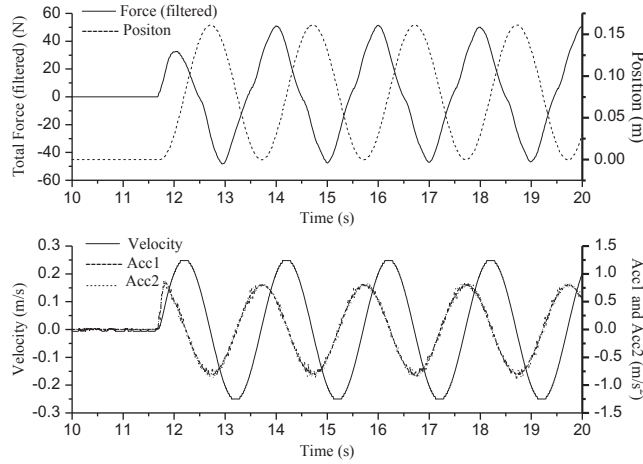


Fig. 4. Typical readings of total force, position, velocity and acceleration of the tested plate, submergence $d = 70$ cm, perforation ratio $\tau = 15.89\%$, period $T = 2.0$ s and $KC = 1.38$.

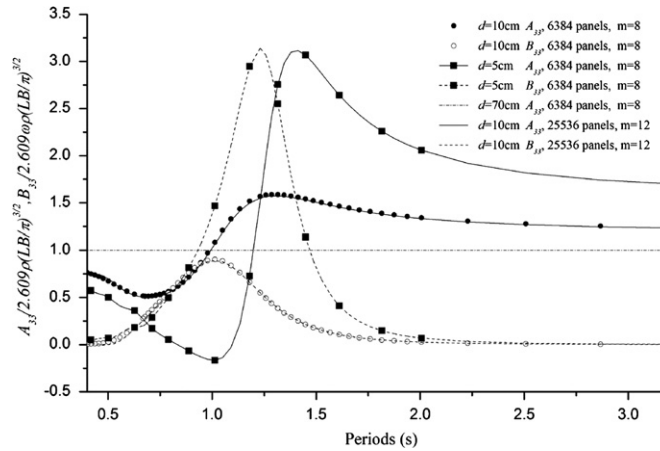


Fig. 5. Numerical nondimensional potential-flow heave added mass and damping for solid plate versus heave period (in seconds) at different submergences ($d = 5, 10$ and 70 cm). Control surface radius $R = 0.5$ m, with 100 concentric-ring elements on S_F , S_R and S_B . Different panel numbers on S_p and expansion-series numbers m in velocity potential are examined.

and damping coefficient were controlled by using Fourier analysis of a selected steady-state force history as shown by Sarphaya and Isaacson (1981).

3.3. Comparisons between experiments and numerical results

Heave added mass and damping coefficients of solid circular discs and plates with varying submergences and heave periods have been numerically examined. For circular discs cases, our numerical results agreed well with the theoretical solutions by Martin and Farina (1997). Convergence studies of cases with submerged plates were performed by changing the panel sizes on the plates S_p , the control surface radius R , the ring-element size, and the expansion-series number m of the velocity potential (see Eq. (12)) used on the control surface S_R , the free surface S_F and the sea-bottom surface S_B . In Fig. 5, we present the heave added mass and damping coefficients of a plate without perforations and of the same length, breadth and thickness as the plate used in the experiments as a function of the heave period. Using 6384 panels on S_p and $m = 8$ in the series expansions on S_F , S_R and S_B agrees well with the results using 25 536 panels on S_p and $m = 12$ in the series expansions on S_F , S_R and S_B for the examined 10 cm submergence case. When the submergence is 70 cm, the calculated added mass and damping of the solid plate is period independent and the same as in infinite fluid. The corresponding damping is zero. The potential-flow calculated heave added mass of the solid plate in the 70 cm submergence case by converged BEM results is $2.609\rho_w(LB/\pi)^{3/2}$. This value is used to nondimensionalize all the experimental and numerical results. The rapid change in added mass and damping as a function of period and with pronounced peak values that occurs for the smallest submergence is a consequence of resonant flow above the plate.

3.3.1. 70 cm submergence case

When the perforated plate is submerged 70 cm below the mean free surface, the distance from the tank bottom to the plate is also 70 cm. In this case the influence of both the top and bottom boundaries is very small. During the experiments no radiated free-surface waves are observed, except for some very small waves generated by the thin rod. However, the wavelengths of the small waves are the order of centimeters and would have negligible effect at the plate. The rods will have an additional non-wave type of disturbance of the flow. Since the flow caused by the plate is nearly vertical at the rods and the rod cross-sectional area is very small relative to the projected area of the plate, we can neglect the latter error source and assume the hydrodynamic damping is only caused by the flow through the openings and around the outer edges of the plate. Experimental nondimensional heave added mass and damping of the perforated plate with four tested heave periods and two different perforation ratios are presented in Figs. 6 and 7. The added mass obtained from the measurements is made nondimensional by our numerically calculated potential-flow heave added mass $2.609\rho_w(LB/\pi)^{3/2}$ of the solid plate without perforations in infinite fluid while damping is made nondimensional by the factor $2.609\omega\rho_w(LB/\pi)^{3/2}$.

In Figs. 6 and 7, a clear KC number dependence of both the nondimensional heave added mass and damping of the perforated plate with the two different perforation ratios are shown. From the second plots in Figs. 6 and 7, the nondimensional damping with two perforation ratios has small sensitivity to the heave period for all the tested KC numbers. The latter fact is a consequence of that wave radiation plays a minor role. However, the added mass is slightly affected by the heave period, which is more prominent in the case of the smaller perforation ratio. The first plot in Fig. 6 which presents the added mass of the plate under four different heave periods at the perforation ratio of 7.945% shows that with the forced heave period increasing the nondimensional added mass is slightly decreasing at all KC numbers. For the perforation ratio 15.89% (first plot in Fig. 7), there is still such tendency but to a lesser degree. The sensitivity of the added mass to the period is, in general, small relative to the dependency on the KC number.

Fig. 8 compares the experimental heave added mass and damping coefficients versus the KC number for the two perforation ratios at a heave period of 2.0 s in the deep submergence case with $d = 70$ cm. Significant influence of the perforation ratio is only observed for the added mass. For the perforation ratio 7.985% the added mass even exceeds the potential-flow solid-plate added mass (shown by the horizontal solid black line) when the KC number is larger than 1.0. The

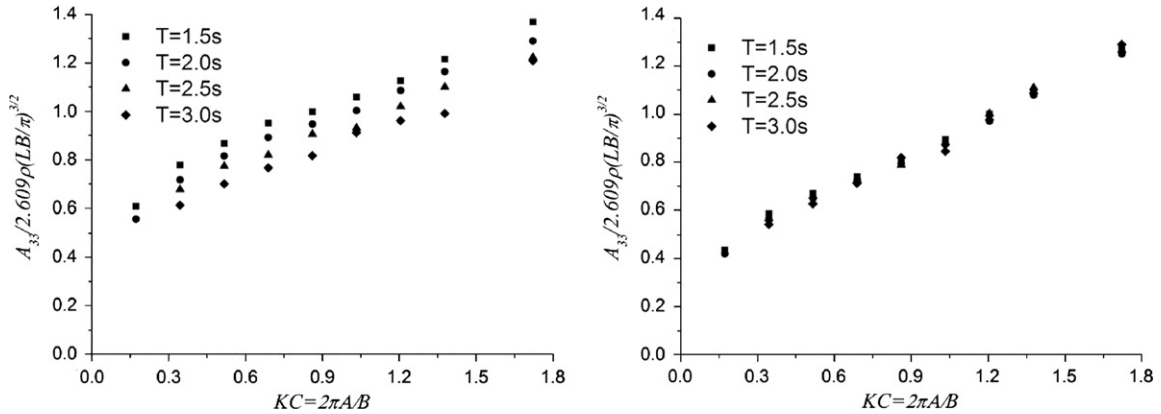


Fig. 6. Experimental heave added mass and damping versus KC number for different heave periods T (in seconds), $d = 70$ cm and $\tau = 7.945\%$.

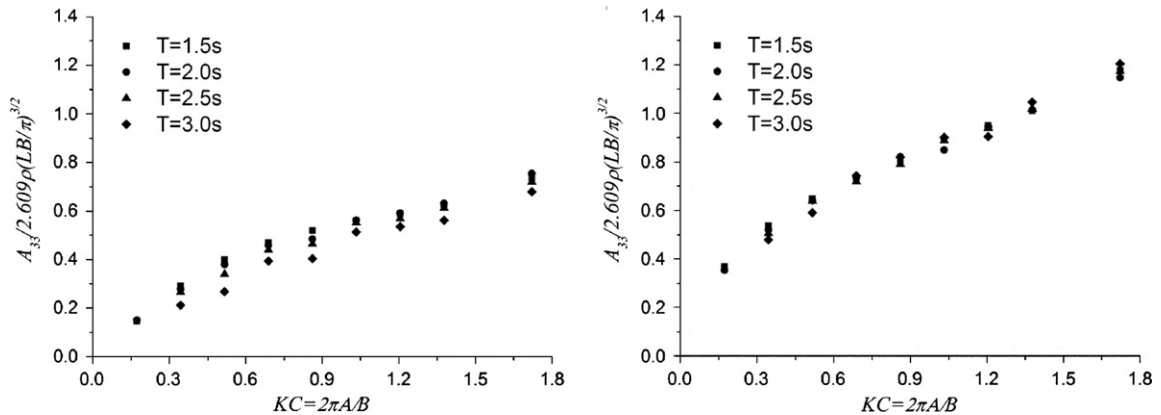


Fig. 7. Experimental heave added mass and damping versus KC number for different heave periods T (in seconds), $d = 70$ cm and $\tau = 15.89\%$.

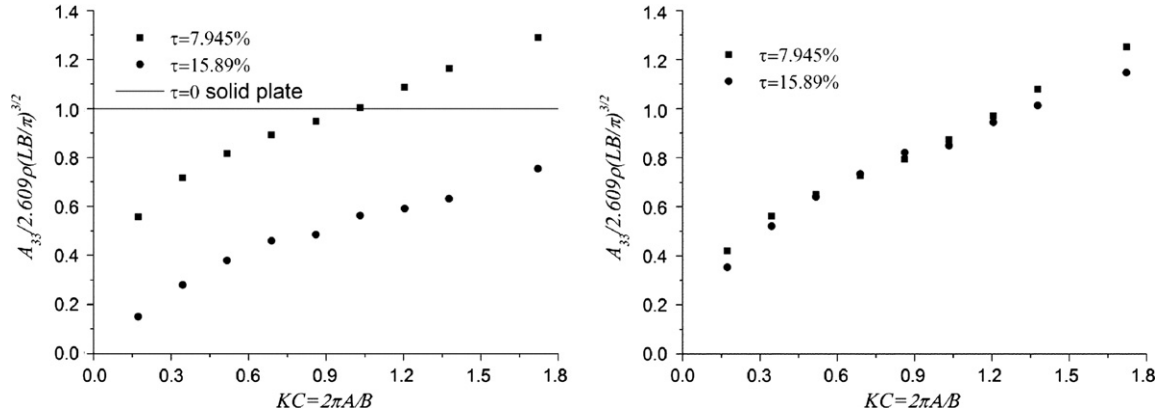


Fig. 8. Experimental heave added mass and damping versus KC number for the two perforation ratios $\tau = 7.945\%$ and 15.89% . $d = 70$ cm and $T = 2.0$ s.

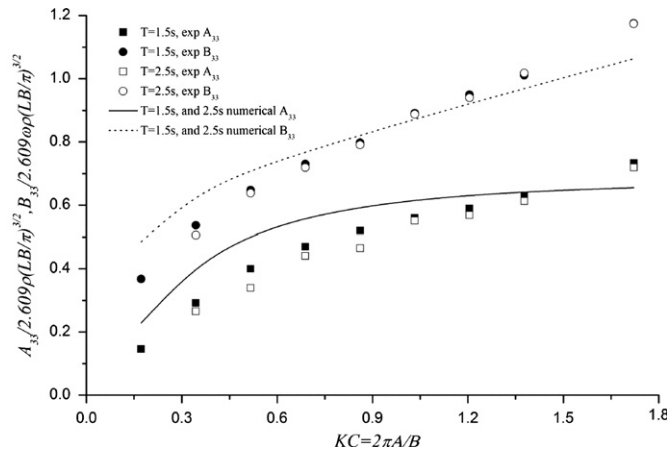


Fig. 9. Nondimensional heave added mass and damping. Comparison between experiments and numerical method. $\mu = 0.9$, $\alpha = 6.0$, $d = 70$ cm, $\tau = 15.89\%$, $T = 1.5$ and 2.5 s.

latter fact is consistent with the experimental results by [Tao and Dray \(2008\)](#). The damping coefficients for the two perforation ratios are almost the same for the mid-range KC numbers from 0.6 to 1.2. A very slight difference only occurred at smaller or larger KC where the smaller perforation ratio gives slightly higher values. So as can be seen in [Fig. 8](#), no extra damping can be gained by increasing the perforation ratio from 7.945% to 15.89%, but the added mass is significantly reduced.

[Fig. 9](#) compares experimental and numerical values of heave added mass and damping for the two heave periods (1.5 s, 2.5 s). A discharge coefficient $\mu = 0.9$ and a proportionality factor $\alpha = 6.0$ in the drag coefficient are used in the numerical calculations. The numerical values are identical at the two periods which is a consequence of small free-surface wave generation. There is generally satisfactory agreement between theory and experiments except for small KC numbers where the numerical values are higher than the experimental values.

[Fig. 10](#) presents comparison between experimental and numerical results with or without the empirical drag term due to flow separation at the outer edges of the plate. The numerical results show the influence of flow through the perforated openings and the flow around the outer edges for both the nondimensional added mass and damping at different KC numbers. The perforation ratio is 15.98% while the period is 1.5 s, with α being 6.0 and μ being 0.9 as an illustration. Because the velocity used in the drag force given by [Eq. \(19\)](#) is the averaged flow velocity along the perforated plate surface, this drag term gives a negative contribution to the added mass, the value of which is the difference between the dotted black line and the solid black line. Clearly this negative contribution increases as the KC number increases. The dotted line with crosses is the hydrodynamic damping predicted by only considering the viscous contribution from the flow through the perforated openings, which reaches its peak at a KC number around 0.4 and intersects with the added mass (dotted black curve) at the nondimensional added mass and damping value around 0.5 and thereafter begins to decrease, and it reaches the lowest value at the highest KC number. The same observation for a 2D perforated plate heaving in infinite fluid was reported by [Molin \(2011\)](#). Similarly, one obtains the damping contribution from the outer edges of the plate by subtracting the dotted line with crosses from the solid line with crosses. This contribution to the total damping becomes more dominant as the KC number increases. When at the largest KC number around 1.7, the contribution to the damping from the outer edges is

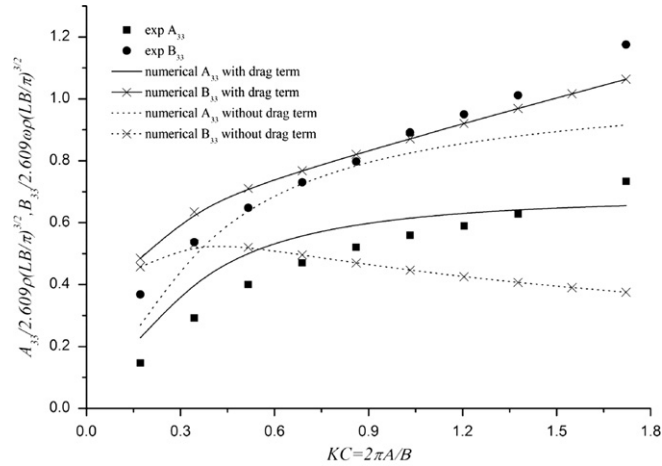


Fig. 10. Nondimensional heave added mass and damping. Comparison between experiments and numerical values with/without drag term due to flow separation at the outer edges versus KC number. $d = 70$ cm, $\tau = 15.89\%$, $T = 1.5$ s, $\alpha = 6.0$ and $\mu = 0.9$

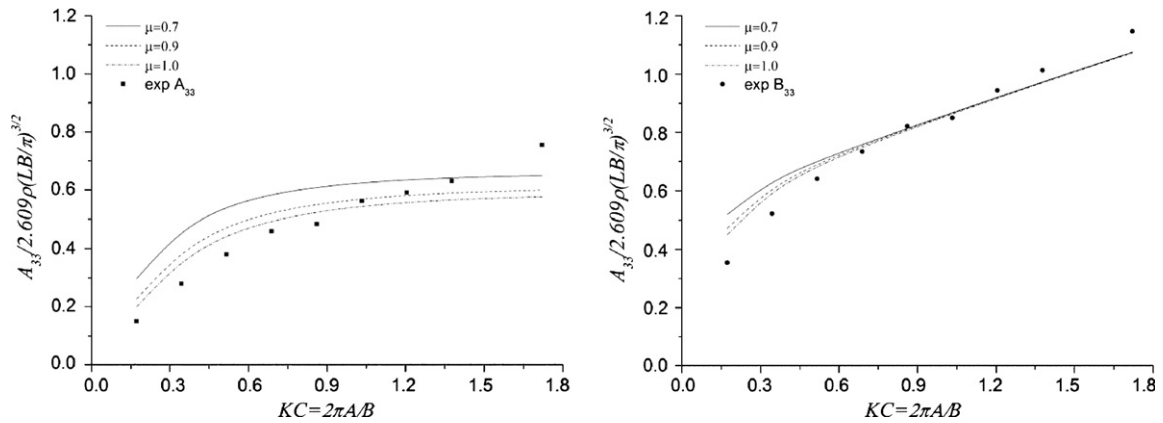


Fig. 11. Numerically predicted heave added mass and damping versus KC number with different discharge coefficients μ . Comparisons with experiments. $d = 70$ cm, $\tau = 15.89\%$, $T = 2.0$ seconds and $\alpha = 6.0$.

almost three times of that from the perforated openings. If we now consider a single perforated plate device as the damping plate for a typical truss spar of diameter 30 m, the heave motion amplitude of the spar is the order of 1 m ($KC \approx 0.2$) according to [Tao and Dray \(2008\)](#). From the damping curve we can see much larger damping can be introduced to the system from the perforation than the vortex shedding due to the edges at the perforation ratio 15.89%.

[Fig. 11](#) illustrates the sensitivity of the heave added mass and damping to the discharge coefficient μ . The proportionality factor α in the drag coefficient is equal to 6.0 and the perforation ratio is 15.98%. For a certain pressure difference between the two perforated plate sides, the larger the discharge coefficient is the easier the water flows through the openings, resulting in a smaller added mass. Further, the added mass curves by the numerical method show a clear tendency of how the added mass reacts to the discharge coefficient. The second plot in [Fig. 11](#) shows how the damping reacts to the discharge coefficient. When the KC number is larger than approximately 0.8, the damping is almost independent of the discharge coefficient.

A flow parameter affecting the flow separation from the outer edges of the plate is the proportionality factor α in the drag coefficient (see Eq. (20)). [Fig. 12](#) shows how the heave added mass and damping vary with α in the deep submergence case. When comparing with the experimental results, $\alpha = 6.0$ gives a better curve fit for damping relative to $\alpha = 8.0$ suggested by [Bearman et al. \(1985\)](#) for a 2D solid plate in infinite fluid. Since the flow will go through the plate perforations as well as around the outer edges, the effect of vortex shedding for the perforated plate will be weaker than the solid plate, which suggests the α value for perforated plate is smaller than the solid plate case. This fact was also noted in the paper by [Molin et al. \(2008\)](#).

3.3.2. 10 cm submergence case

[Fig. 13](#) presents experimental heave added mass and damping as a function of the KC number for the two studied submergences $d = 10$ cm and $d = 70$ cm. The perforation ratio is 7.945% and the forcing period is 2.0 s. The experimental

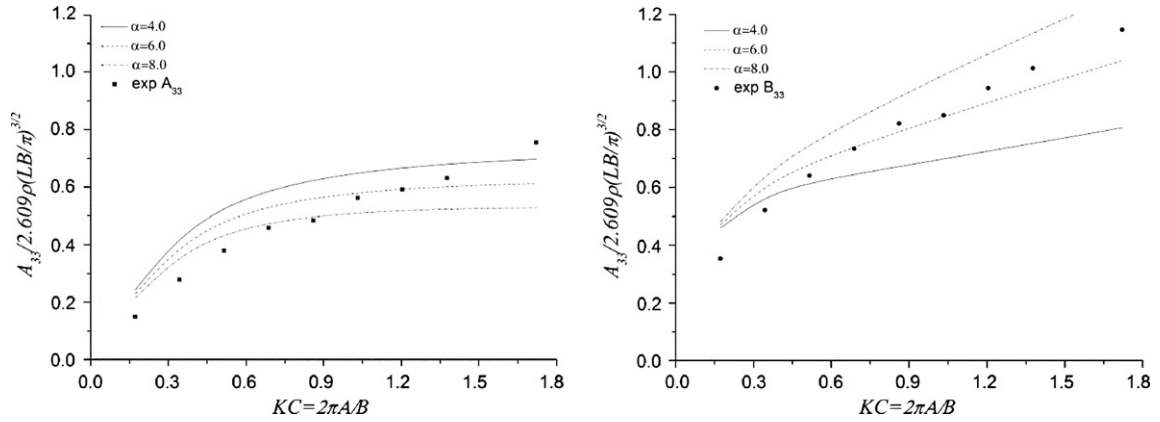


Fig. 12. Numerically predicted heave added mass and damping versus KC number with different proportionality factor α in the drag coefficient. $d = 70$ cm, $\tau = 15.89\%$, $T = 2.0$ s and $\mu = 0.9$.

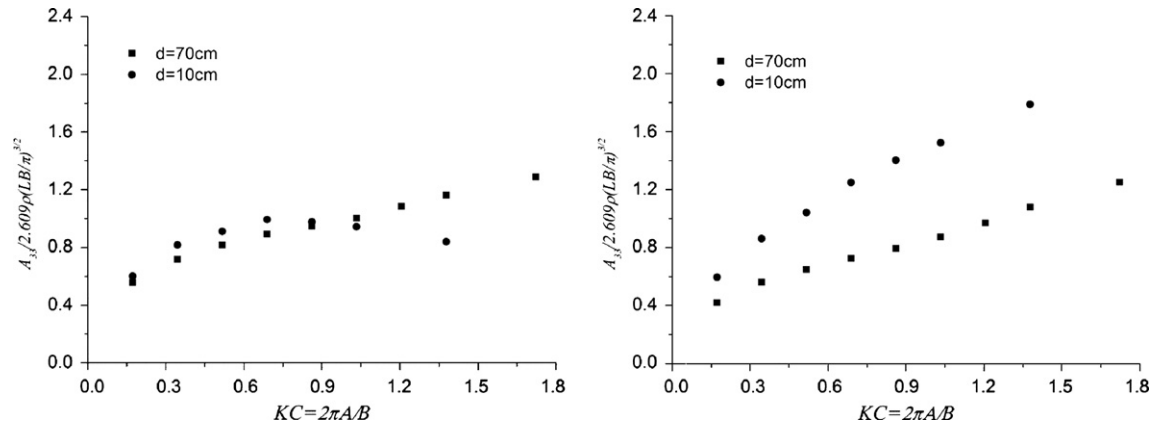


Fig. 13. Nondimensional heave added mass and damping versus the KC number with different submergences $d = 10$ cm and $d = 70$ cm. $\tau = 7.945\%$, $T = 2.0$ s.

damping values at the smaller submergence are clearly higher than for the larger submergence for a given KC number. An obvious reason is the increased importance of free-surface wave generation with decreasing submergence. The added mass is monotonically increasing with the KC number for the larger submergence while it has a maximum value at $KC \approx 0.8$ – 1.0 for the smaller submergence.

Fig. 14 presents experimental added mass and damping as a function of the KC number for the larger perforation ratio and the smaller submergence with forcing period 2.0 s. Similarly as for the smaller perforation ratio the damping increases with the KC number and the added mass has a maximum value at $KC \sim 1.0$. Comparisons are also made with numerical values by varying the proportionality factor α in the drag coefficient from 8.0 to 12.0. A higher value of α is needed in the smaller submergence case than in the larger submergence case to get the best fit to the experimental values. The most appropriate α -value will depend on the oscillation period as also noted by Molin et al. (2008) for a heaving disk. The heave amplitude is not small relative to the mean submergence for the higher KC numbers. For instance, when $KC = 1.40$, the forced heave amplitude-to-mean submergence ratio is 0.8. The latter fact implies nonlinear effects. For instance, we ought to numerically satisfy the body boundary conditions and evaluate the pressure at the instantaneous plate position instead of on the mean position. Further, nonlinear free-surface conditions are likely to matter, in particular when resonant water motion happens above the plate for smaller submergences.

Fig. 15 shows the numerically predicted nondimensional heave added mass and damping coefficients of the perforated plate versus the forced heave period. The predictions are compared with experimental results and numerical values for the solid plate case. The parameters $\alpha = 9.0$ and $\mu = 0.9$ have been used. The perforation ratio is 15.89% and the forced heave amplitude is 4 cm for all the periods. The figure clearly shows that the perforation greatly reduces the added mass relative to the solid plate case at all the heaving periods of our interest. The damping peak for the solid plate occurs at a period around 1.0 s. From the experimental results, the largest damping occurs at a period around 1.6 s. Although the α -value should not be the same for all the heave periods, the numerical model gives reasonable predictions.

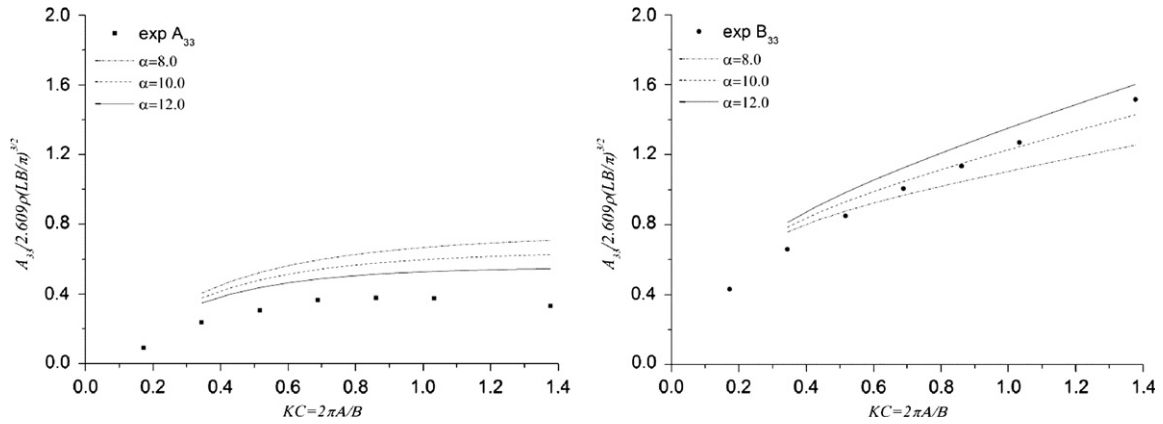


Fig. 14. Nondimensional heave added mass and damping versus KC number with different proportionality factor α in the drag coefficient. $d = 10$ cm, $\tau = 15.89\%$, $T = 2.0$ s and $\mu = 0.9$.

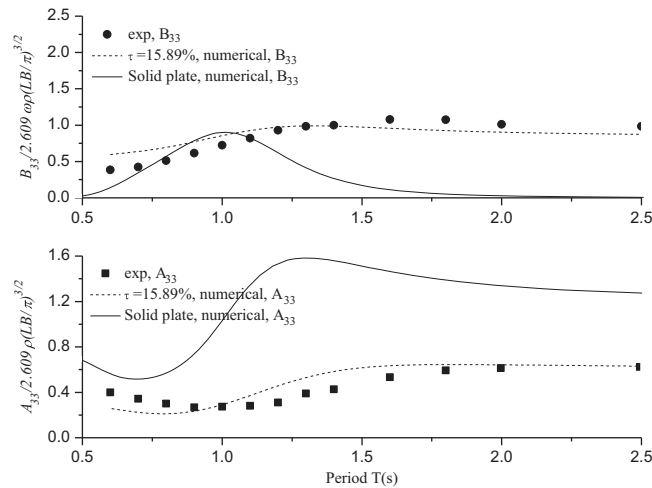


Fig. 15. Nondimensional heave added mass and damping versus oscillation period T in seconds. Comparison between numerical and experimental results. $d = 10$ cm, $\tau = 15.89\%$, $\mu = 0.9\alpha = 9.0$ and $KC = 0.69$.

4. Conclusions

This paper examines numerically and experimentally the heave added mass and damping of horizontally submerged and perforated rectangular plates. The effects of the perforation ratio, plate submergence, forcing period and heave amplitude (KC number) are studied. Based on that, the following conclusions can be drawn.

When the perforated plate is deeply submerged, both the numerically predicted and experimentally measured added mass and damping indicate stronger dependence on the KC number than on the frequency. The numerical KC-dependent hydrodynamic coefficients are due to the nonlinear relation between the pressure drop/loss and the relative cross-flow velocity at the perforations. The perforation of the plate reduces the added mass for the studied perforation ratios at all KC numbers. When the KC number is small, the damping is mainly caused by the flow through the perforated openings. However, when the KC number is larger, the flow separation from the outer edges of the plate dominates the hydrodynamic damping and reduces the added mass. The numerical method gives satisfactory predictions for the hydrodynamic coefficients in the studied KC range between 0.17 and 1.72.

When the perforated plate heaves close to the free-surface, experimental results show that the drag coefficient is greatly affected by the free surface. The mechanism of the frequency dependence of the proportionality factor α in the drag coefficient for the perforated plate close to the free surface needs to be further studied. In order to achieve best curve fit with the experimental results, the boundary integral equation should be solved at the instantaneous position of the plate, together with the nonlinear free-surface conditions, especially when the forced heave amplitude of the plate is the order of the submergence.

Generally speaking, the potential-flow theory based BEM with pressure loss/drop model and empirical drag coefficients provides a reasonable option to study the nonlinear properties and the averaged effects caused by the perforation of the

plate. The advantage of using BEM is that it can be applied to a variety of plate shapes and also can be used to study the hydrodynamic behavior of a perforated plate together with other bodies of interests at the same time.

Cases where a perforated plate penetrates the free surface or has a very small submergence relative to what we have examined need further studies.

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Reference

- Bearman, P.W., Downie, M.J., Graham, J.M.R., Obasaju, E.D., 1985. Forces on cylinders in viscous oscillatory flow at low Keulegan–Carpenter numbers. *Journal of Fluid Mechanics* 154, 337–365.
- Blevins, R.D., 1984. *Applied Fluid Mechanics Handbook*. Van Nostrand Reinhold Company.
- Chua, K.H., Clelland, D., Huang, S., Sworn, A., 2005. Model experiments of hydrodynamic forces on heave plates. In: *Proceedings of the 24th International Conference on Offshore Mechanics and Arctic Engineering*, Halkidiki, Greece.
- Chwang, A.T., Chan, A.T., 1998. Interaction between porous media and wave motion. *Annual Review of Fluid Mechanics* 30, 53–84.
- Downie, M.J., Wang, J., Graham, J.M.R., 2000. The effectiveness of porous damping devices. In: *Proceedings of the 10th International Offshore and Polar Engineering Conference*, Seattle, USA, pp. 418–425.
- He, H., Troesch, A.W., Perlin, M., 2008. Hydrodynamics of damping plates at small KC numbers. *IUTAM Symposium on Fluid–Structure Interaction in Ocean Engineering*, vol. 8; 93–104.
- Hess, J.L., Smith, A.M.O., 1967. Calculation of potential flow about arbitrary bodies. *Progress in Aerospace Sciences* 8, 1–138.
- Lake, M., He, H., Troesch, A.W., Perlin, M., Thiagrajan, K.P., 2000. Hydrodynamic coefficient estimation for TLP and SPAR structures. *Journal of Offshore Mechanics and Arctic Engineering* 122 (2), 118–124.
- Martin, P.A., Farina, L., 1997. Radiation of water waves by a heaving submerged horizontal disc. *Journal of Fluid Mechanics* 337, 365–379.
- Molin, B., 1992. Motion damping by slotted structures. In: van den Boom, HJJ (Ed.), *Hydrodynamics: Computations, Model Tests and Reality. Developments in Marine Technology*, vol. 10; 1992, pp. 297–303.
- Molin, B., 1993. A potential flow model for the drag of shrouded cylinders. *Journal of Fluids and Structures* 7, 29–38.
- Molin, B., 2001. On the added mass and damping of periodic arrays of fully or partially porous disks. *Journal of Fluids and Structures* 15, 275–290.
- Molin, B., 2011. Hydrodynamic modeling of perforated structures. *Applied Ocean Research* 33 (1), 1–11.
- Molin, B., Fourest, J.M., 1992. Numerical modeling of progressive wave absorbers. In: *Proceedings of the Seventh International Workshop on Water waves and Floating bodies*, Val de Reuil, France, pp. 119–203.
- Molin, B., Korobkin, A.A., 2001. Water entry of a perforated wedge. In: *Proceedings of the 16th International Workshop on Water Waves and Floating Bodies*, Hiroshima, Japan.
- Molin, B., Legras, J.L., 1990. Hydrodynamic modeling of the Roseau tower stabilizer. In: *Proceedings of the Ninth International Conference on Offshore Mechanics and Arctic Engineering*, Houston, USA, pp. 329–336.
- Molin, B., Nielsen, F.G., 2004. Heave added mass and damping of a perforated disk below the free surface. In: *Proceedings of the 19th International Workshop on Water Waves and Floating Bodies*, Cortona, Italy.
- Molin, B., Remy, F., Rippol, T., 2008. Experimental study of the heave added mass and damping of solid and perforated disks close to the free surface, Maritime Industry, Ocean Engineering and Coastal Resources Taylor & Francis 879–887.
- Prislin, I., Blevins, R.D., Halkyard, J.E., 1998. Viscous damping and added mass of solid square plates. In: *Proceedings of the 17th International Conference on Offshore Mechanics and Arctic Engineering*, Lisbon, Portugal.
- Sandvik, P.C., Solaas, F., Nielsen, F.G., 2006. Hydrodynamic forces on ventilated structures. In: *Proceedings of the 16th International Offshore and Polar Engineering Conference*, San Francisco, California, USA.
- Sarpkaya, T., Isaacson, M., 1981. *Mechanics of Wave Forces on Offshore Structures*. Van Nostrand Reinhold Company.
- Tao, L., Dray, D., 2008. Hydrodynamic performance of solid and porous heave plates. *Ocean Engineering* 35, 1006–1014.